

Chapter 54

Community Ecology

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: A Sense of Community

- A biological **community** is an assemblage of populations of various species living close enough for potential interaction. *All life / all populations in an area.*
- Ecologists call relationships *between species* in a community ***interspecific interactions***.
- Interspecific interactions can affect the survival and reproduction of each species. *Effects can be positive (+), negative (–), or no effect (0).*
- Examples: competition, predation, herbivory, and symbiosis (parasitism, mutualism, commensalism).

Competition

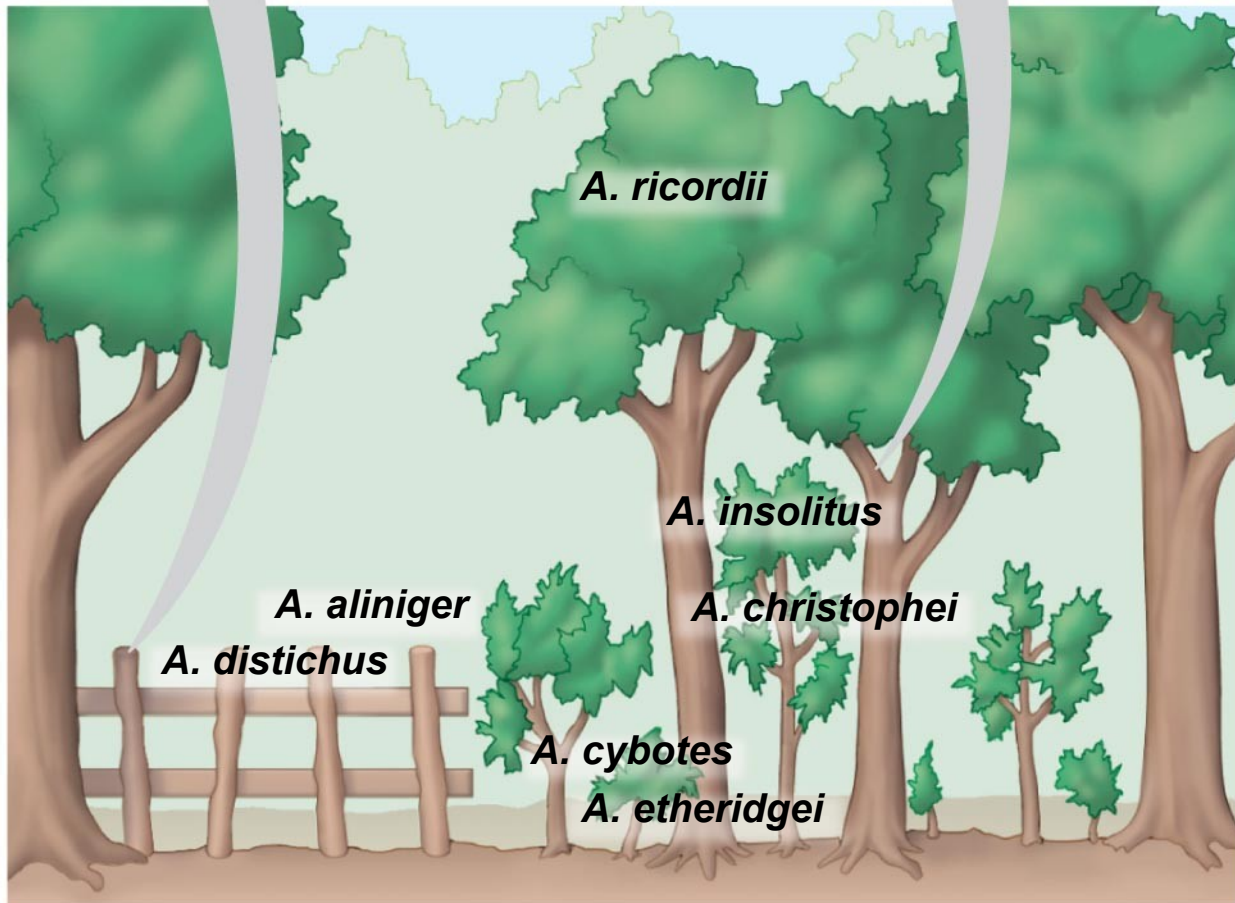
- **Interspecific competition** (–/– interaction) occurs when **different species compete** for a resource in short supply.
- Strong competition can lead to **competitive exclusion**, local elimination of a competing species.
- The ***competitive exclusion principle*** states that two species competing for the same limiting resources cannot coexist in the same place = **1 species per niche**.

Ecological Niches

- *The total of a species' use of biotic and abiotic resources is called the species' **ecological niche**.*
- An ecological niche can also be thought of as **an organism's ecological role**.
- Ecologically similar species can coexist in a community if there are one or more significant differences in their niches.
- **Resource partitioning** is differentiation of ecological niches; **enables similar species to coexist in a community**.

A. Lizard species perches on fences and other sunny surfaces.

B. lizard species usually perches on shady branches.



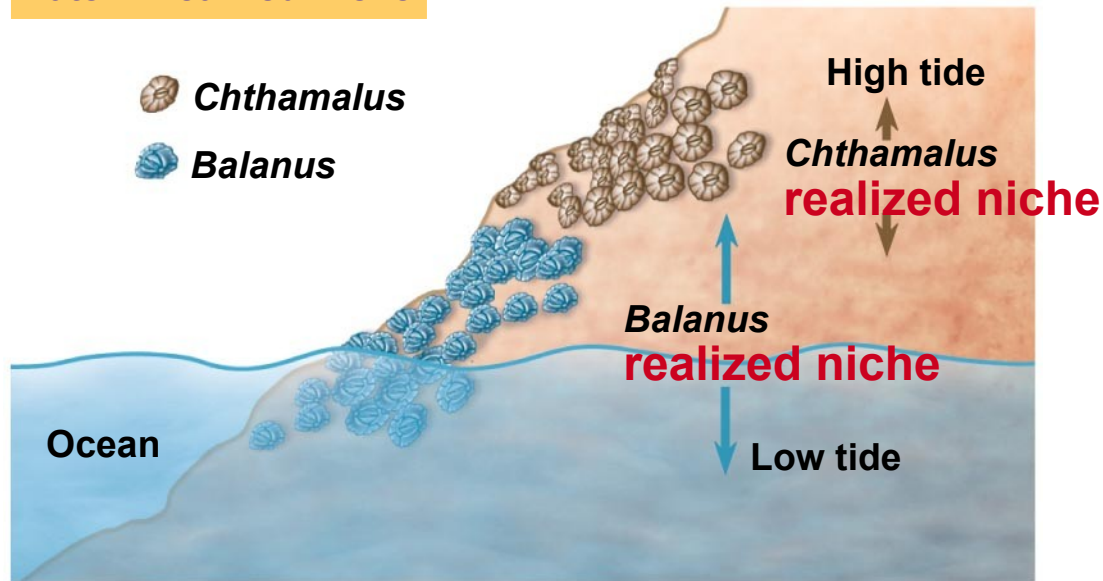
Resource partitioning is differentiation of ecological niches, enabling similar species to coexist in a community

Interspecific => Competition Between Species: Can Lead to Resource Partitioning

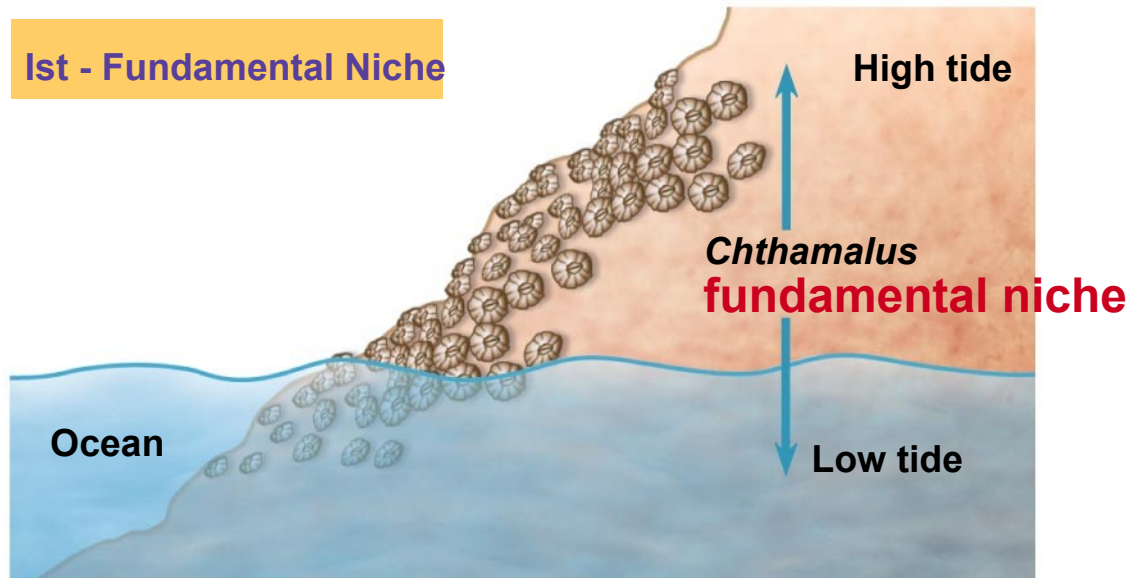
- As a result of **interspecific competition**, a species' fundamental niche may differ from its **realized niche --> the niche it occupies** after resource partitioning.

How a species' niche can be influenced by interspecific competition?

Later - Realized Niche



1st - Fundamental Niche

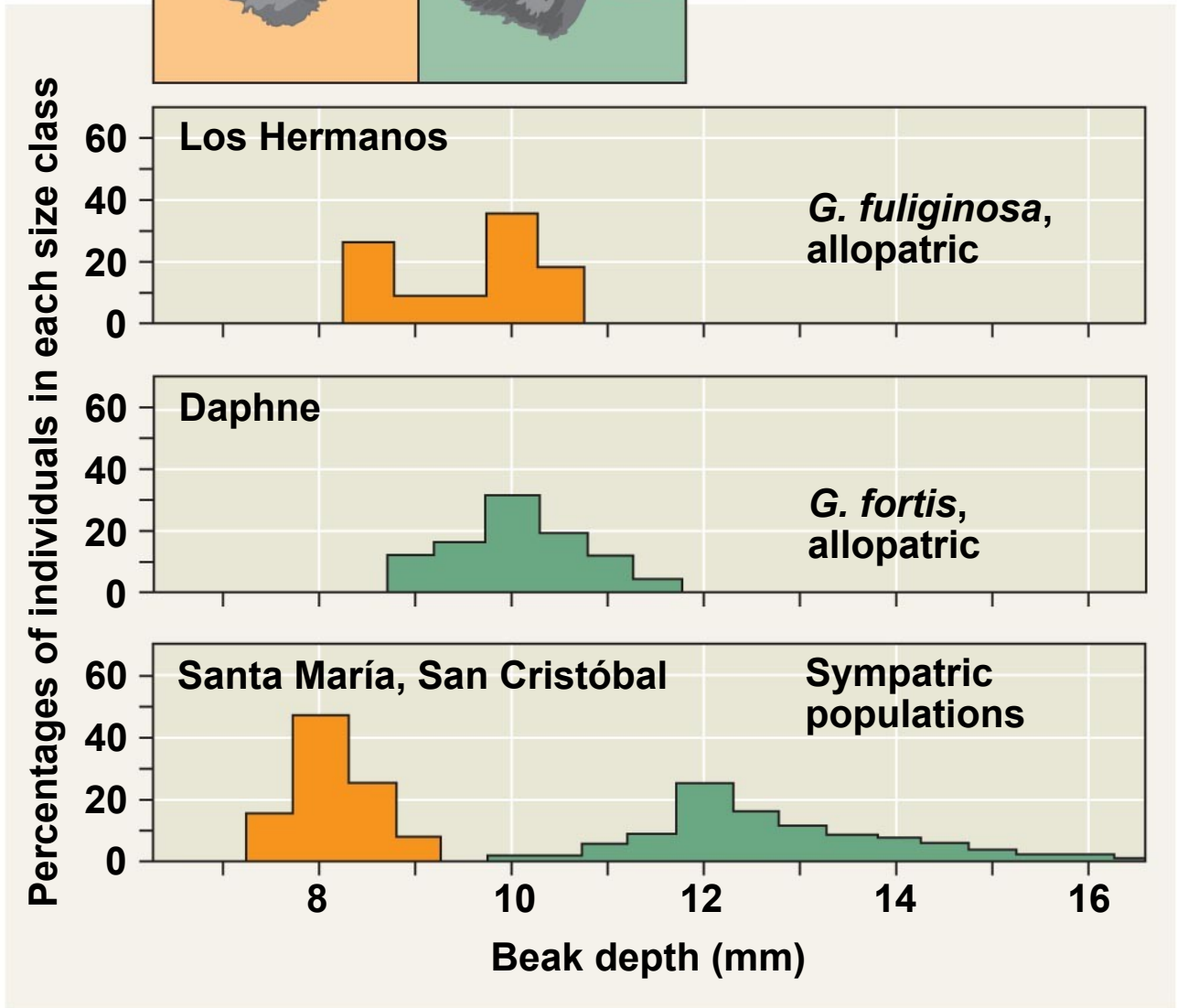
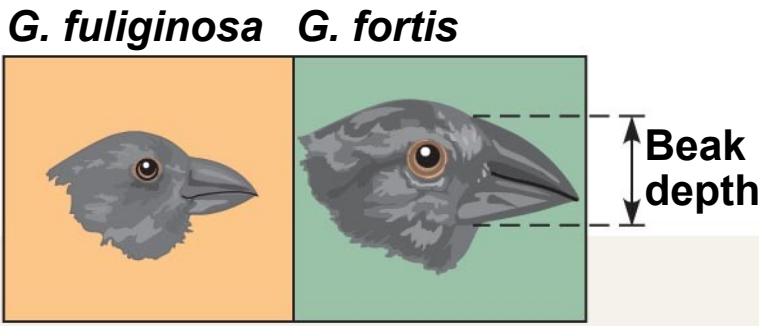


Character Displacement

- **Character displacement** is a tendency for characteristics / particular traits to be more divergent in sympatric populations of two species than in allopatric populations of the same two species.
- An example is variation in beak size between populations of two species of Galápagos finches.

Character displacement:

Indirect Evidence of Past Competition



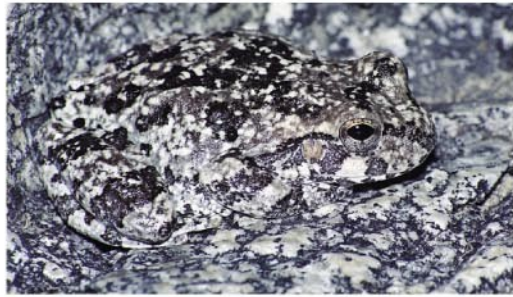
Predation

- **Predation** (+/– interaction) refers to interaction where one species, the predator, kills and eats the other, the prey.
- Some **feeding adaptations** of predators are claws, teeth, fangs, stingers, and poison.
- **Prey** display various **defensive adaptations**: such as **behavior** and **coloration**.

Prey: Defensive Adaptations

- ***Behavioral*** defenses include hiding, fleeing, forming herds or schools, self-defense, and alarm calls.
- Animals also have ***morphological and physiological defense adaptations:***
- ***Cryptic coloration*** = *camouflage*, makes prey difficult to spot.
- ***Aposematic coloration:*** Animals with effective chemical defense / *poison* / often exhibit *bright warning coloration*. Predators are particularly cautious in dealing with prey that display such coloration.

- (a) **Cryptic coloration**
▶ Canyon tree frog



- (b) **Aposematic coloration**
▶ Poison dart frog



- (c) **Batesian mimicry**: A harmless species mimics a harmful one.



▶ Hawkmoth larva

▼ Green parrot snake



- (d) **Müllerian mimicry**: Two “yuck” unpalatable species mimic each other.



▶ Cuckoo bee

▼ Yellow jacket



Mimicry = “Look-alikes” Defense

- In some cases, a prey species may gain significant **protection** by mimicking the appearance of another species:
- In ***Batesian mimicry***, a harmless species mimics an unpalatable or harmful model... ***One is a “pretender.”***
- In ***Müllerian mimicry***, two or more unpalatable species resemble each other... ***BOTH are “yuck.”***

Herbivory: *Herbivores = Plant Predators*

- **Herbivory** (+/– interaction) refers to an interaction in which an herbivore eats parts of a plant or alga.
- It has led to evolution of ***plant defenses against herbivores: secondary compounds*** = are ***chemical defenses***; and ***mechanical defenses which are often osmoregulated.***

Symbiosis: + + + 0 + -

- ***Symbiosis*** is a *dependency relationship* where two or more species live in direct and intimate contact with one another. The relationship is generally based on one or some combination of the following ***benefits***:
 - *Nutrition (food, water)*
 - *Protection*
 - *Reproduction*

Parasitism + -

- In **parasitism** (+/– interaction), one organism, the **parasite**, derives nourishment from another organism, its **host**, which is harmed in the process.
- **Endoparasites** = parasites that live within the body of their host.
- **Ectoparasites** = parasites that live on the external surface of a host.
- Many parasites have a complex life cycle involving a number of hosts.
- Some parasites change the behavior of the host to increase their own fitness (reproduce more offspring).

Mutualism ++

- Mutualistic symbiosis, or **mutualism** (+/+ interaction), is an interspecific interaction that **benefits both species**.
- A mutualism can be:
 - **Obligate** = *MUST* where one species cannot survive without the other.
 - **Facultative** = *OPTIONAL* where both species can survive alone.

Commensalism + 0

- In **commensalism** (+/0 interaction), one species benefits and the other is apparently unaffected.
- Commensal interactions are hard to document in nature because any close association likely affects both species.

A possible example of commensalism between cattle egrets (birds) and water buffalo: The Birds eat insects disturbed by the Buffalo as they move.



Dominant and keystone species exert strong controls on **community structure**

- A few species in a community often exert strong control on that community's structure.
- Two fundamental features of community structure = **species diversity** and **feeding relationships**.

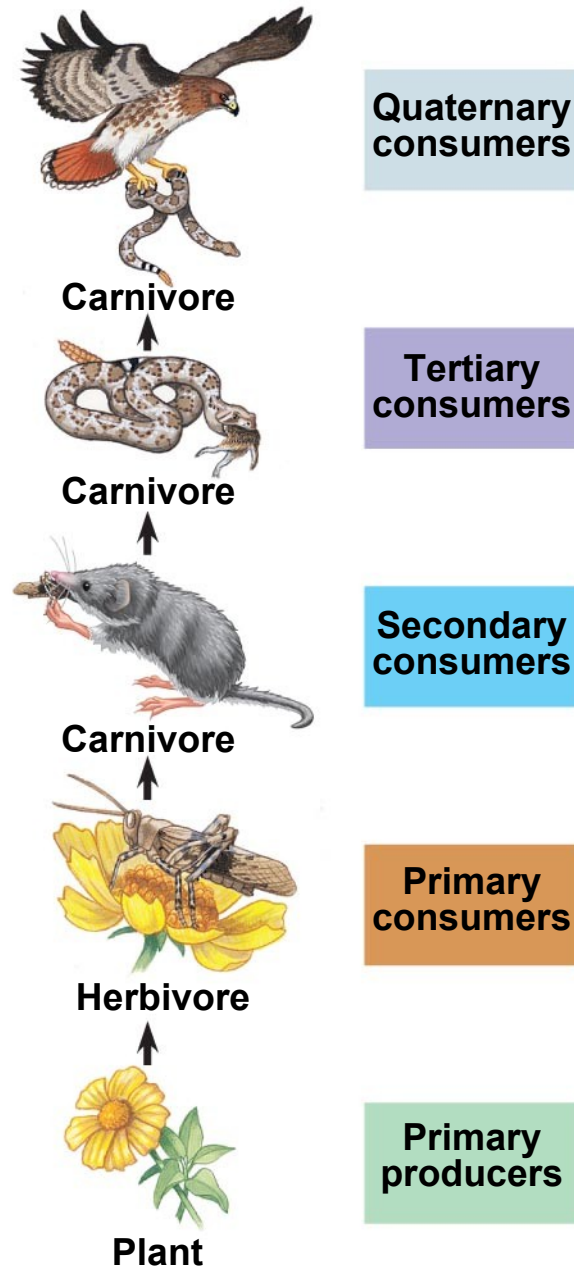
Species Diversity

- **Species diversity** of a community is the **variety of organisms** that make up the community.
- It has two components: species richness and relative abundance.
- **Species richness** is the total number of different species in the community.
- **Relative abundance** is the proportion each species represents of the total individuals in the community.

Trophic Structure = a key factor in community dynamics

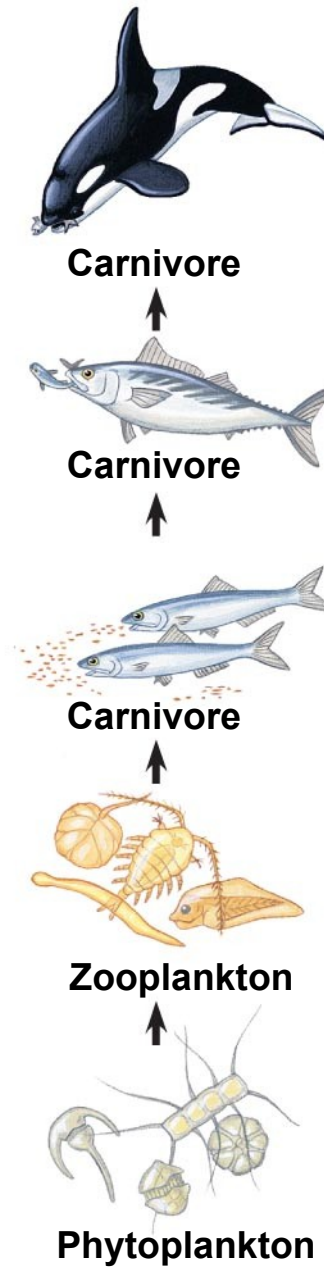
- **Trophic structure** is the **feeding relationships** between organisms in a community.
- **Food chains** link trophic levels from producers to top carnivores.
- A **food web** is a branching food chain with complex trophic interactions.
- Species may play a role at more than one trophic level.
- Food chains in a food web are usually only a few links long. WHY?

Terrestrial and Marine Food Chains



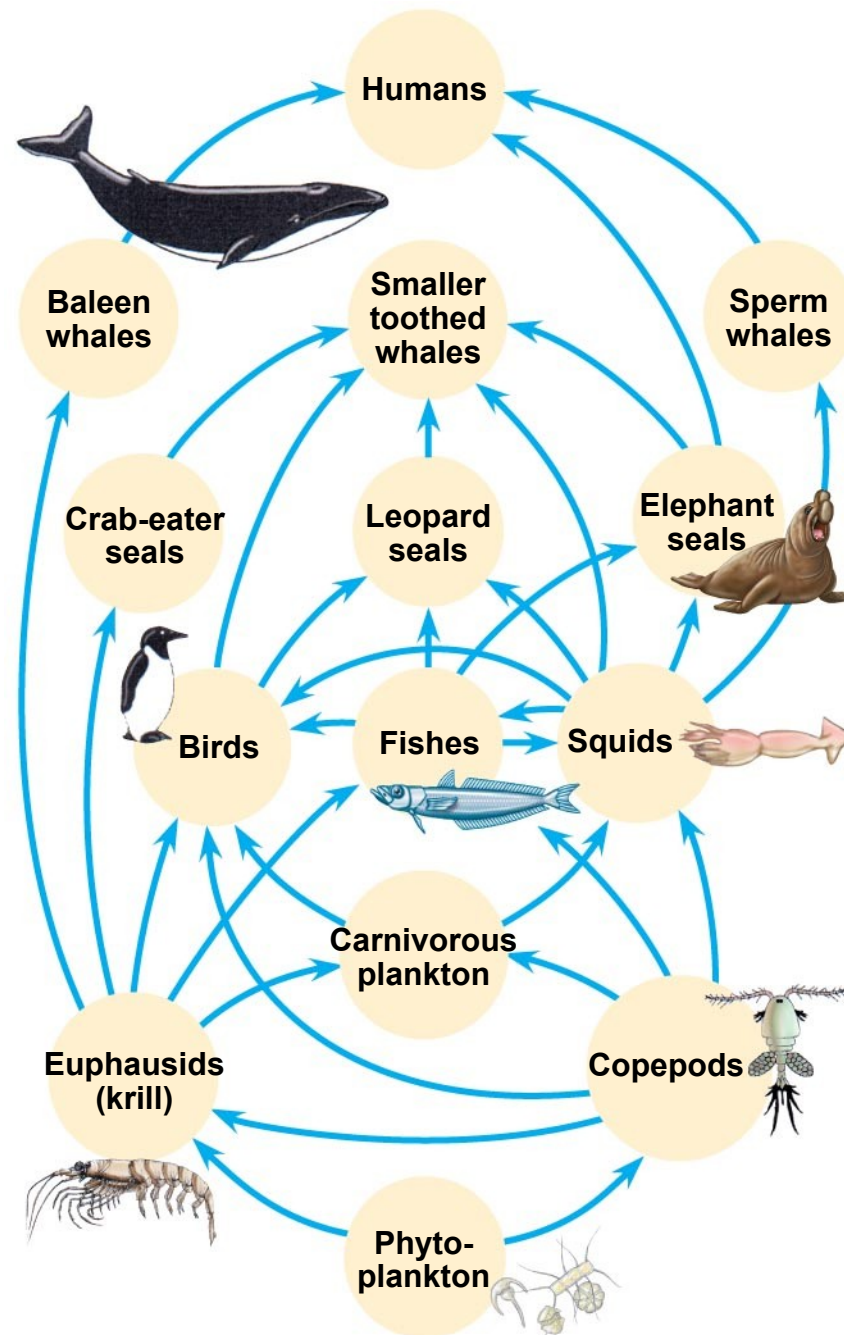
A **terrestrial** food chain

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A **marine** food chain

An Antarctic Marine Food Web



Limits on Food Chain Length

- Food chains in food webs are usually only a few links long.
- Two hypotheses attempt to explain food chain length: the *energetic hypothesis* and the *dynamic stability hypothesis*.
- The *energetic hypothesis* suggests that *length is limited by inefficient energy transfer*.
- The **dynamic stability hypothesis** proposes that long food chains are less stable than short ones.
- Most data support the energetic hypothesis.

Species with a Large Impact

- Certain species have a very large impact on community structure. Such species are *highly abundant* **OR** play a *pivotal role* in community dynamics.
- ***Dominant species*** = those that are most abundant or have the *highest biomass*.
- Biomass is the total mass of all individuals in a population. Dominant species exert powerful control over the occurrence and distribution of other species.

Invasive species, typically introduced to a new environment by humans, often lack predators or disease pathogens. *Invasive species disrupt ecosystem dynamics.* They frequently out-compete / displace native populations.

Keystone Species

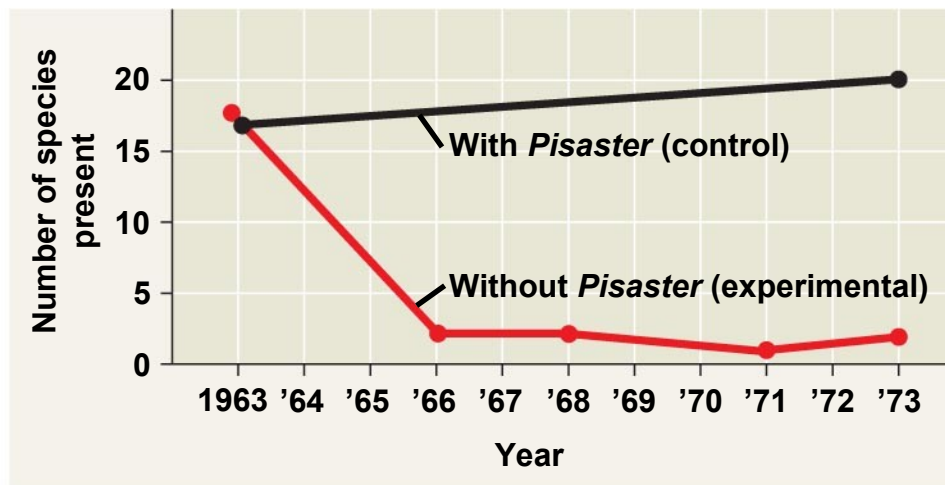
- ***Keystone species*** exert *strong control on a community by their ecological roles*, or niches.
- In contrast to dominant species, they are not necessarily abundant in a community.
- Field studies of sea stars exhibit their role as a keystone species in intertidal communities.
- Sea otter populations and their predation shows how otters affect ocean communities. Sea otters are keystone predators in the North Pacific.

Seastar are keystone predators. They are key in preserving species diversity in their ecosystem.

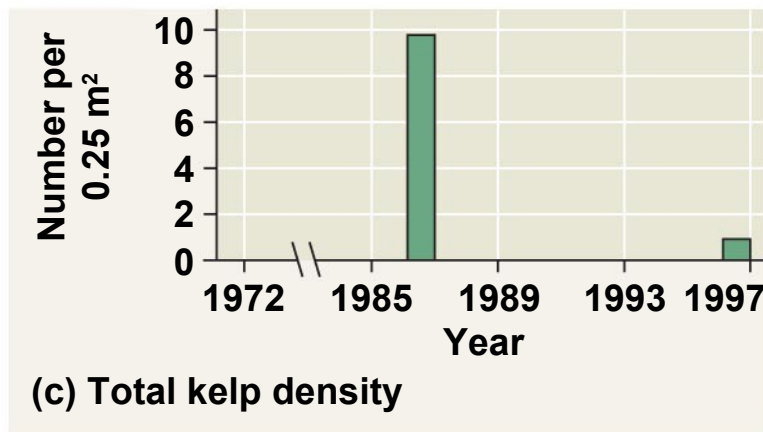
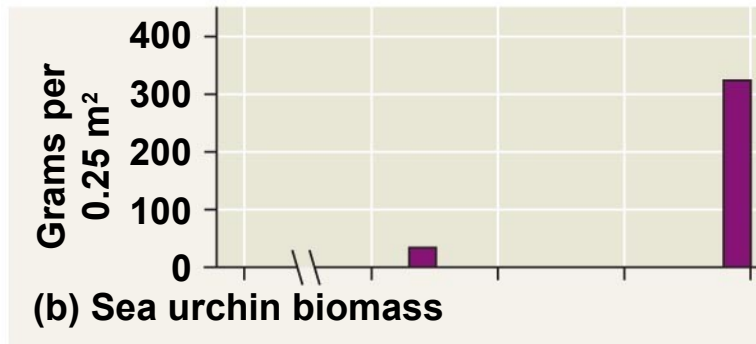
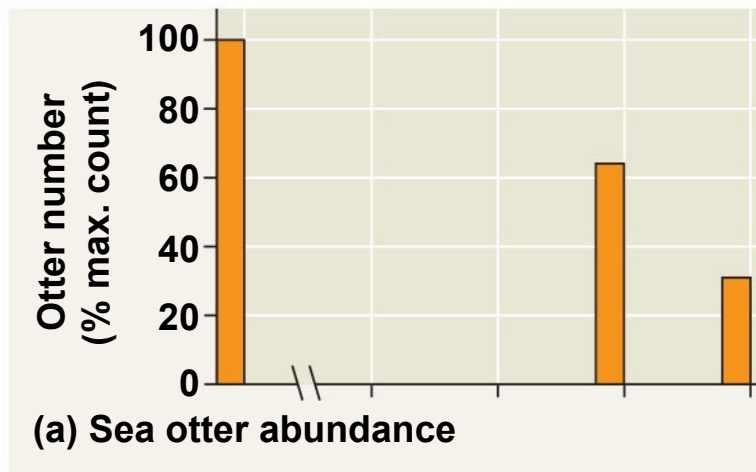
EXPERIMENT



RESULTS



Sea otters are keystone predators in the North Pacific



Food chain

Foundation Species (Ecosystem “Engineers”)

- Foundation species (ecosystem “engineers”) cause physical changes in the environment that affect community structure.
- For example, beaver dams can transform landscapes on a very large scale.
- Some *foundation species* act as **facilitators** that have positive effects on survival and reproduction of some other species in the community.

Beavers are a Foundation Species = ecosystem“engineers”



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Bottom-Up and Top-Down Controls

- The **bottom-up model** of community organization proposes a unidirectional influence from lower to higher trophic levels.
- In this case, presence or absence of mineral nutrients determines community structure, including abundance of primary producers.
- The **top-down model**, also called the trophic cascade model, proposes that control comes from the trophic level above.
- In this case, predators control herbivores, which in turn control primary producers.

Disturbance influences species diversity and composition

- Pollution can affect community dynamics.
- **Biomanipulation** can help restore polluted communities. *Bio remediation* is an effective strategy to *restore* polluted and damaged areas.
- Decades ago, most ecologists favored the view that communities are in a state of equilibrium.
- Recent evidence of change has led to a **nonequilibrium model**, which describes communities as constantly changing after being buffeted by **disturbances**.

Characterizing Disturbance

- A disturbance is an event that changes a community, removes organisms from it, and alters resource availability.
- Fire is a *significant large scale disturbance* in most terrestrial ecosystems. It is often a necessity in some communities.
- The *intermediate disturbance hypothesis* suggests that moderate levels of disturbance can foster greater diversity than either high or low levels of disturbance.

The large-scale fire in Yellowstone National Park in 1988 demonstrated that communities can often respond very rapidly to a massive disturbance.



(a) Soon after fire



(b) One year after fire

Ecological Succession

Ecological succession is the *sequence of community and ecosystem changes* after a disturbance, over time.

- ***Primary succession*** occurs where *no soil* exists when succession begins. *Pioneer organisms*, such as *lichen*, are the foundation of the community and *soil building*.
- ***Secondary succession*** begins in an area where *soil remains* after a disturbance / *disaster* such as fire or field abandonment.

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- Early-arriving species and later-arriving species may be linked in one of three processes:
 - Early arrivals may facilitate appearance of later species by making the environment favorable
 - They may inhibit establishment of later species
 - They may tolerate later species but have no impact on their establishment
 - *Glacier retreating -- predictable pattern of ecological succession ...*



1 Pioneer stage = soil builders / fireweed dominant



2 **Dryas stage** **grasses and shrubs**



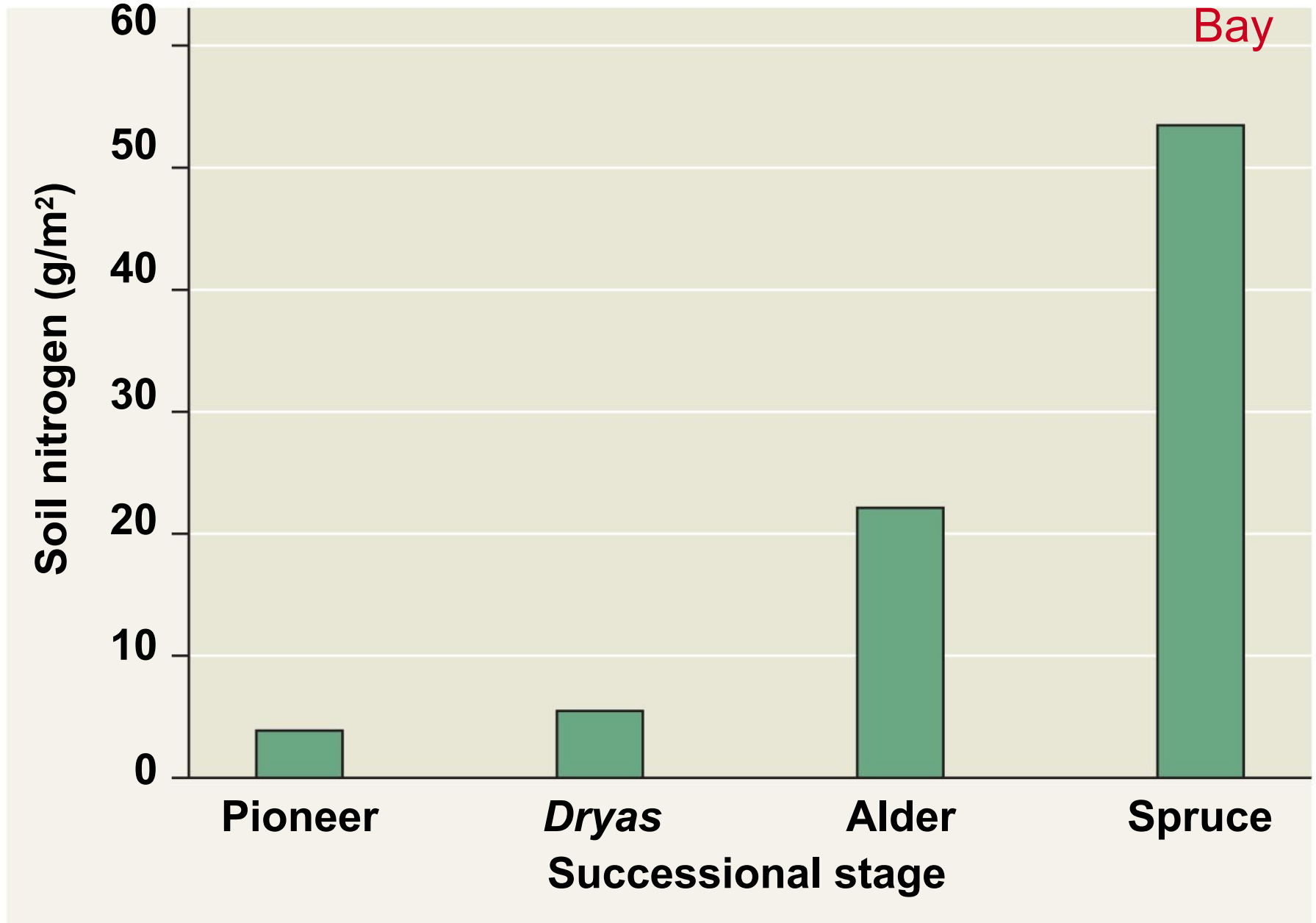
3 Alder stage: **trees** and shrub



4 Spruce stage = Climax Community STABLE

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- *Succession is the result of changes induced by the vegetation itself.*
 - On the glacial moraines, vegetation lowers the ***soil pH*** and increases ***soil nitrogen*** content.

Changes in soil nitrogen content during succession at Glacier Bay



Human Disturbance

- Humans have the greatest impact on biological communities worldwide. Human disturbance to communities usually reduces species diversity.
- Humans also prevent some naturally occurring disturbances, which can be important to community structure.

Disturbance of the ocean floor by trawling



Biogeographic factors affect community biodiversity

- ***Latitude*** and ***area*** are two key factors that affect a ***community's species diversity***.
- Species richness generally declines along an equatorial-polar gradient and is especially great in the tropics.
- Two key factors in ***equatorial-polar gradients*** of species richness are probably ***evolutionary history*** and ***climate***.
- The greater age of tropical environments may account for the greater species richness.

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- **Climate** is likely the primary cause of the latitudinal gradient in biodiversity.
 - Two main **climatic factors** correlated with biodiversity are **solar energy and water** availability. They can be considered together by measuring a community's rate of **evapotranspiration**.
 - **Evapotranspiration** is evaporation of water from soil plus transpiration of water from plants.

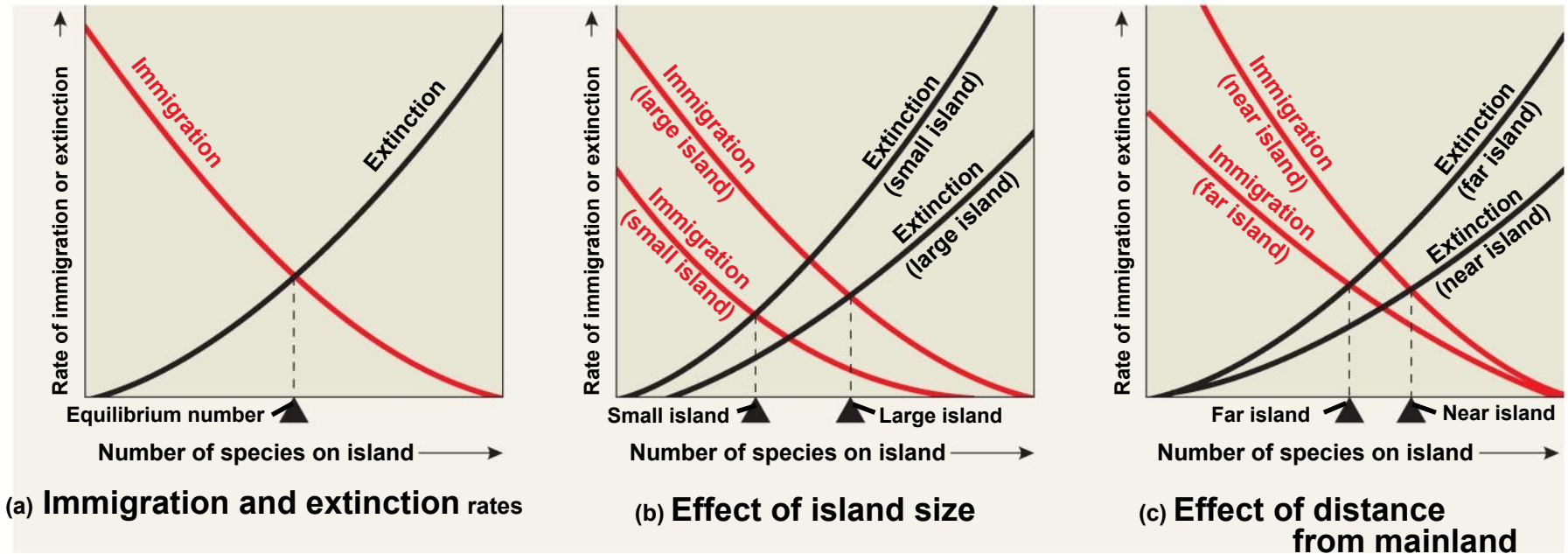
Area Effects

- The **species-area curve** quantifies the idea that, all other factors being equal, **a larger geographic area has more species.**
- A species-area curve of North American breeding birds supports this idea.

Island Equilibrium Model

- **Species richness on islands** depends on island **size, distance from the mainland, immigration, and extinction.**
- The equilibrium model of island biogeography maintains that species richness on an ecological island levels off at a dynamic equilibrium point.
- Studies of species richness on the Galápagos Islands support the prediction that species richness increases with island size.

The equilibrium model of island biogeography



Community ecology is useful for understanding pathogen life cycles and controlling human disease

- Ecological communities are universally affected by **pathogens**, which include **disease-causing microorganisms, viruses, viroids, and prions**.
- Pathogens can alter community structure quickly and extensively.
- For example, coral reef communities are being decimated by white-band disease.

White-band disease on coral is destroying the reef.



Community Ecology and Zoonotic Diseases

- Human activities are transporting pathogens around the world at unprecedented rates.
- Community ecology is needed to help study and combat them.
- **Zoonotic** pathogens have been transferred from other animals to humans.
- The transfer of **pathogens** can be direct or through an intermediate species called a **vector**.
- Many of today's emerging human diseases are zoonotic. Avian flu is a highly contagious virus of birds.

Interspecific Interaction	Description
Competition (−/−)	Two or more species compete for a resource that is in short supply. The competitive exclusion principle states that two species cannot coexist in the same community if their niches (ecological roles) are identical.
Predation (+/−)	One species, the predator, kills and eats the other, the prey. Predation has led to diverse adaptations, including mimicry.
Herbivory (+/−)	An herbivore eats part of a plant or alga. Plants have various chemical and mechanical defenses against herbivory, and herbivores have specialized adaptations for feeding.
Symbiosis	Individuals of two or more species live in close contact with one another. Symbiosis includes parasitism, mutualism, and commensalism.
Parasitism (+/−)	The parasite derives its nourishment from a second organism, its host, which is harmed.
Mutualism (+/+)	Both species benefit from the interaction.
Commensalism (+/0)	One species benefits from the interaction, while the other is unaffected by it.

You should now be able to:

1. Distinguish between the following sets of terms: competition, predation, herbivory, symbiosis; fundamental and realized niche; cryptic and aposematic coloration; Batesian mimicry and Müllerian mimicry; parasitism, mutualism, and commensalism; endoparasites and ectoparasites; species richness and relative abundance; food chain and food web; primary and secondary succession.

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2. Define an ecological niche and explain the competitive exclusion principle in terms of the niche concept.
 3. Explain how dominant and keystone species exert strong control on community structure.
 4. Distinguish between bottom-up and top-down community organization.
 5. Describe and explain the intermediate disturbance hypothesis.

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6. Explain why species richness declines along an equatorial-polar gradient.
 7. Define zoonotic pathogens and explain, with an example, how they may be controlled.